UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Low-Cost Fiber Optic Interrogator and Energy Harvesting Technologies for Power Grid Sensing Applications

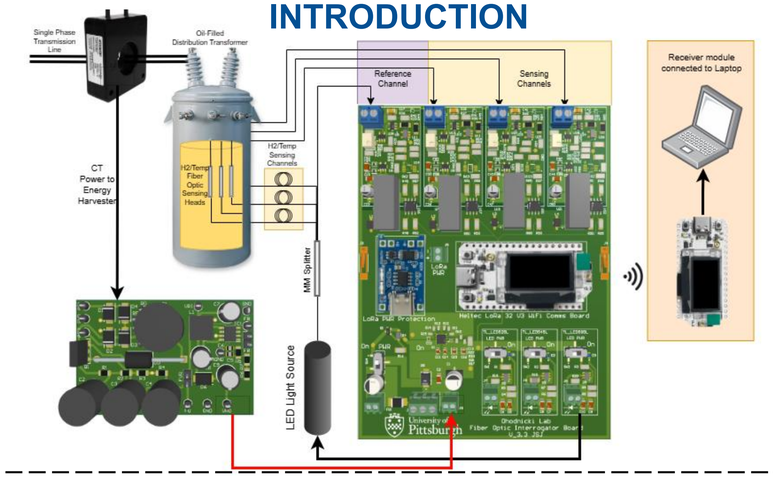
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INTRODUCTION



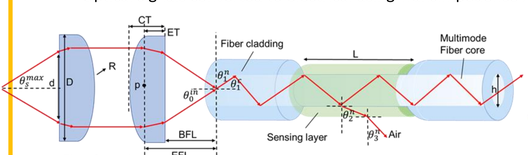
Increased electrical demand from the grid has caused in additional strain on pre-existing distributive transformer technologies, resulting in a reduction of their lifespan and increasing their chance of failure. Monitoring state-of-health (SoH) on these systems allows for measurement of their current life cycle and detect when significant degradation is present. Using fiber optics offers a low cost, reliable approach that can be integrated into pre-existing systems with its small footprint and the sensor's high electromagnetic tolerance. This work encapsulates a multi-channel interrogator that interfaces with multiple fiber optic sensors that measure real-time temperature and hydrogen gas concentrations. Transmitted optic signals from sensors are sampled across each channel and sent over Long Range (LoRa) wireless radio for remote sensing. This work also includes development of an energy harvesting system, which can harvest power line electromagnetic waves to provide localize power for the interrogator unit. Prototypes of both systems have been demonstrated with ongoing validation efforts.

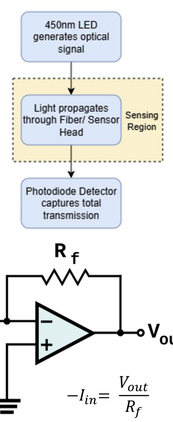
METHODOLOGIES

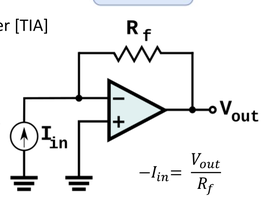
Sensor Implementation and Application

The interrogator system handles both driving and measuring of the incident optical signal, capturing total signal transmittance through the fiber and sensing heads to each respective channel.

- LEDs are driven at continuous operating point for selected wavelength
- Sensor head design is unique for sensing application:
  - Gold Palladium [AdPd] – H2 Sensing
  - Gold embedded plasmonic nanoparticles [$AuSIO_2$] – Temperature Sensing
- Optical signals converted to electrical using Transimpedence Amplifier [TIA]







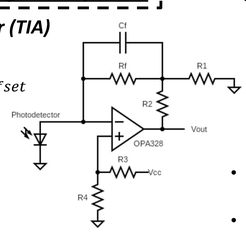
METHODOLOGIES

Mutli-Channel Interrogator System

Transimpedence Amplifier (TIA)

$$V_{out} = I_{pd}\left(\frac{R_f R_2}{R_1}\right) + V_{offset}$$

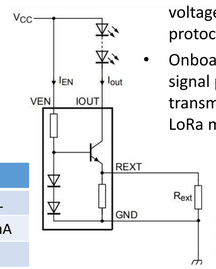
$$V_{offset} = V_{cc}\left(\frac{R_4}{R_3 + R_4}\right)$$



- Transmitted light to photodetector generates electron carriers to diode junction, causing current to flow
- Transimpedance Amplifier configured with 1G feedback, allowing for 1nA of current sensitivity in linear region
- Capacitor feedback provides stability by limiting fluctuations; Offset allows for vertical shifting of IV Curve

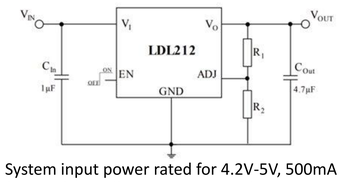
LED Driver Circuitry

- Power to LED Sources are regulated with NCR421U LED Driver IC for continuous power delivery
- Interrogator board is fitted with three circuits for varied wavelength delivery and testing



LED continuous ratings		
LED450LW	LED525L	LED545L
2.8V/150mA	3.0V/30mA	3.8V/50mA
90mW output	4mW output	2.4mW

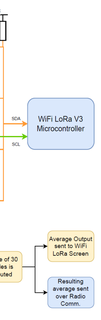
System Linear Regulator



- System input power rated for 4.2V-5V, 500mA from either wall power or energy harvester
- Input power regulated with LDL212 configurable Linear regulator – steps to 3.3V across board
- Decoupling capacitors to handle transient events

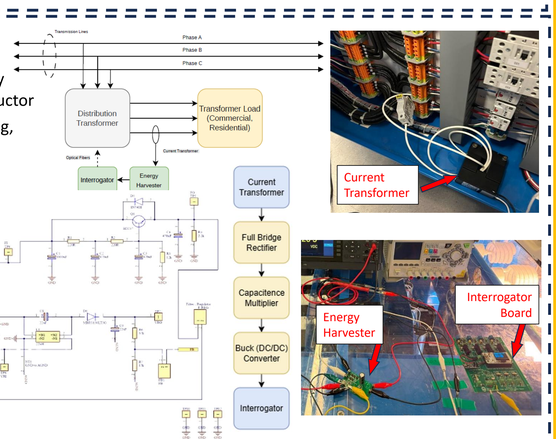
Sampling and Wireless Transmission

- Indexes system ADCs (ADS1115) to capture voltage reading over I2C protocol
- Onboard MC assembles signal packet, sends transmission to Receiver LoRa module



Energy Harvester System

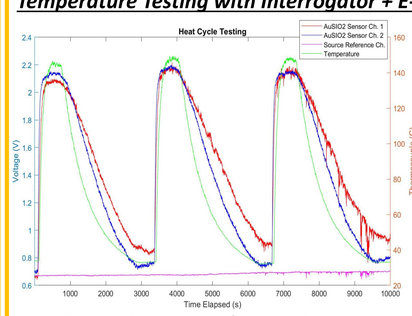
- E-Harvester system harvests EMF energy using Current Transformer (CT) on conductor
- Regulates CT energy by rectifying, storing, and stepping down DC voltage to 5V
- Built to withstand up to 1A continuous power delivery

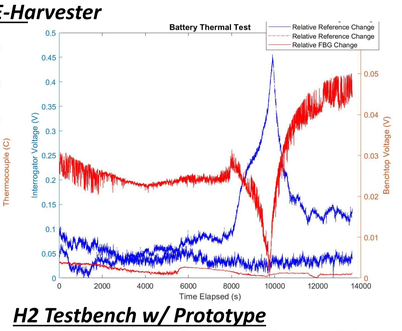


System Ratings	
CT Configuration	20:4 Turn Ratio
E-Harvester max input	35Vac/1A
E-Harvester output	5V/500mA
Interrogator req. power	4.2-5V/450mA

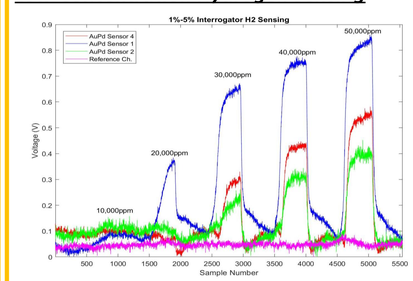
PRELIMINARY RESULTS & PERSPECTIVES

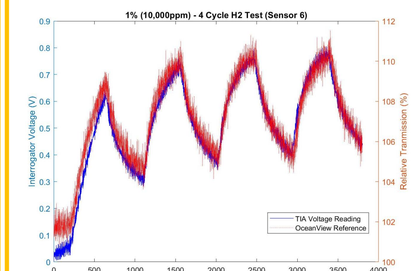
Temperature Testing with Interrogator + E-Harvester



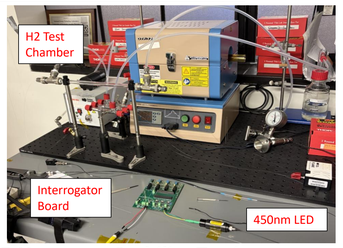


Low-Concentration Hydrogen Sensing

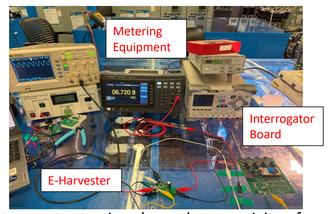




H2 Testbench w/ Prototype



E-Harvester Stability Testing



- Interrogator testing showed responsivity of multi-cycle heat rise: ~1.3mV/°C (±1.1mV)
- High gain TIA (1G ohm) showed responsivity of 400mV for 1% (10,000ppm) H2 in test chamber (blue); distinct responses with 1%-5% test
- Working to improve signal response and stability

ACKNOWLEDGEMENT

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U.S. DEPARTMENT of ENERGY

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